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► **B** REGULATION (EU) 2023/1804 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL
of 13 September 2023
on the deployment of alternative fuels infrastructure, and repealing Directive 2014/94/EU
(Text with EEA relevance)
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**REGULATION (EU) 2023/1804 OF THE EUROPEAN
PARLIAMENT AND OF THE COUNCIL**

of 13 September 2023

**on the deployment of alternative fuels infrastructure, and repealing
Directive 2014/94/EU**

(Text with EEA relevance)

Article 1

Subject matter

1. This Regulation establishes mandatory national targets leading to the deployment of sufficient alternative fuels infrastructure in the Union for road vehicles, trains, vessels and stationary aircraft. It lays down common technical specifications and requirements on user information, data provision and payment requirements for alternative fuels infrastructure.

2. This Regulation also establishes rules for the national policy frameworks referred to in Article 14 to be adopted by the Member States, including rules for the deployment of alternative fuels infrastructure in areas where no mandatory Union-wide targets are set and for reporting on the deployment of such infrastructure.

3. This Regulation establishes a reporting mechanism to encourage cooperation and ensures robust tracking of progress. The reporting mechanism shall take the form of a structured, transparent and iterative process taking place between the Commission and Member States for the purpose of finalising the national policy frameworks, taking into account existing local and regional strategies for the deployment of alternative fuels infrastructure, and their subsequent implementation and corresponding Commission action to support the coherent and more rapid deployment of alternative fuels infrastructure in the Member States.

Article 2

Definitions

For the purposes of this Regulation, the following definitions apply:

- (1) ‘accessibility of data’ means the possibility to request and obtain data at any time in a machine readable format;
- (2) ‘ad hoc price’ means the price charged by the operator of a recharging or refuelling point to an end user for recharging or refuelling on an ad hoc basis;
- (3) ‘along the TEN-T road network’ means:
 - (a) with regard to electric recharging stations: that they are located on the TEN-T road network or within 3 km driving distance from the nearest exit of a TEN-T road; and
 - (b) with regard to hydrogen refuelling stations: that they are located on the TEN-T road network or within 10 km driving distance from the nearest exit of a TEN-T road;

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- (4) ‘alternative fuels’ means fuels or power sources which serve, at least partly, as a substitute for fossil oil sources in the energy used for transport and which have the potential to contribute to its decarbonisation and enhance the environmental performance of the transport sector, including:
- (a) ‘alternative fuels for zero-emission vehicles, trains, vessels or aircraft’:
 - electricity,
 - hydrogen,
 - ammonia,
 - (b) ‘renewable fuels’:
 - biomass fuels, including biogas, and biofuels as defined in Article 2, points (27), (28) and (33), respectively, of Directive (EU) 2018/2001,
 - synthetic and paraffinic fuels, including ammonia, produced from renewable energy,
 - (c) ‘non-renewable alternative fuels and transitional fossil fuels’:
 - natural gas in gaseous form (compressed natural gas (CNG)) and liquefied form (liquefied natural gas (LNG)),
 - liquefied petroleum gas (LPG),
 - synthetic and paraffinic fuels produced from non-renewable energy;
- (5) ‘aircraft contact stand’ means a stand in a designated area of the airport apron equipped with a passenger boarding bridge;
- (6) ‘aircraft remote stand’ means a stand in a designated area of the airport apron not equipped with a passenger boarding bridge;
- (7) ‘airport of the TEN-T core network or airport of the TEN-T comprehensive network’ means an airport as listed and categorised in Annex II to Regulation (EU) No 1315/2013;
- (8) ‘automatic authentication’ means the authentication of a vehicle at a recharging point through the recharging connector or telematics;
- (9) ‘availability of data’ means the existence of data in a digital machine-readable format;
- (10) ‘battery electric vehicle’ means an electric vehicle that runs exclusively on the electric motor, with no secondary source of propulsion;
- (11) ‘bi-directional recharging’ means a smart recharging operation where the direction of the electricity flow can be reversed, allowing that electricity flows from the battery to the recharging point it is connected to;

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- (12) ‘connector’ means the physical interface between the recharging or refuelling point and the vehicle through which the fuel or electric energy is exchanged;
- (13) ‘commercial air transport’ means ‘commercial air transport’ as defined in Article 3, point (24), of Regulation (EU) 2018/1139 of the European Parliament and of the Council ⁽¹⁾;
- (14) ‘container ship’ means a ship designed exclusively for the carriage of containers in holds and on deck;
- (15) ‘contract-based payment’ means a payment for a recharging or refuelling service from the end user to a mobility service provider on the basis of a contract concluded between that end user and that mobility service provider;
- (16) ‘data user’ means any public authority, road authority, road operator, recharging and refuelling point operator, research or non-governmental organisation, mobility service provider, e-roaming platform, digital map provider or any other entity interested in using data to provide information, create services or perform research or analysis on alternative fuels infrastructure;
- (17) ‘digitally-connected recharging point’ means a recharging point that can send and receive information in real time, communicate bi-directionally with the electricity grid and the electric vehicle, and that can be remotely monitored and controlled, including in order to start and stop the recharging session and to measure electricity flows;
- (18) ‘distribution system operator’ means a ‘distribution system operator’ as defined in Article 2, point (29), of Directive (EU) 2019/944;
- (19) ‘distributor’ means a ‘distributor’ as defined in Article 3, point (43), of Regulation (EU) 2018/858 of the European Parliament and of the Council ⁽²⁾;
- (20) ‘dynamic data’ means data that change often or on a regular basis;
- (21) ‘electric road system’ means a physical installation along a road for the transfer of electricity to an electric vehicle while the vehicle is in motion;
- (22) ‘electric vehicle’ means a motor vehicle equipped with a powertrain containing at least one non-peripheral electric machine as energy converter with an electric rechargeable energy storage system, which can be recharged externally;

⁽¹⁾ Regulation (EU) 2018/1139 of the European Parliament and of the Council of 4 July 2018 on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency, and amending Regulations (EC) No 2111/2005, (EC) No 1008/2008, (EU) No 996/2010, (EU) No 376/2014 and Directives 2014/30/EU and 2014/53/EU of the European Parliament and of the Council, and repealing Regulations (EC) No 552/2004 and (EC) No 216/2008 of the European Parliament and of the Council and Council Regulation (EEC) No 3922/91 (OJ L 212, 22.8.2018, p. 1).

⁽²⁾ Regulation (EU) 2018/858 of the European Parliament and of the Council of 30 May 2018 on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles, amending Regulations (EC) No 715/2007 and (EC) No 595/2009 and repealing Directive 2007/46/EC (OJ L 151, 14.6.2018, p. 1).

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- (23) ‘electricity supply to stationary aircraft’ means the supply of electricity through a standardised fixed or mobile interface to an aircraft when stationed at an aircraft contact stand or at an aircraft remote stand;
- (24) ‘end user’ means a natural or legal person purchasing an alternative fuel for direct use in a vehicle;
- (25) ‘e-roaming’ means the exchange of data and payments between the operator of a recharging or refuelling point and a mobility service provider from which an end user purchases a recharging or refuelling service;
- (26) ‘e-roaming platform’ means a platform connecting market actors, notably mobility service providers and operators of recharging or refuelling points, to enable the provision of services between them, including e-roaming;
- (27) ‘European standard’ means a ‘European standard’ as defined in Article 2, point (1)(b), of Regulation (EU) No 1025/2012;
- (28) ‘general aviation’ means all civil aviation operations other than scheduled air services and non-scheduled air transport operations for remuneration or hire;
- (29) ‘gross tonnage’ (GT) means ‘gross tonnage’ as defined in Article 3, point (e), of Regulation (EU) 2015/757 of the European Parliament and of the Council ⁽³⁾;
- (30) ‘heavy-duty vehicle’ means a Category M₂ motor vehicle as described in Article 4(1), point (a)(ii), a Category M₃ motor vehicle as described in Article 4(1), point (a)(iii), a Category N₂ motor vehicle as described in Article 4(1), point (b)(ii) or a Category N₃ motor vehicle as described in Article 4(1), point (b)(iii), of Regulation (EU) 2018/858;
- (31) ‘high-power recharging point’ means a recharging point with a power output of more than 22 kW for the transfer of electricity to an electric vehicle;
- (32) ‘high-speed passenger craft’ means a ‘high-speed craft’ as defined in Regulation 1 of Chapter X of the International Convention for the Safety of Life at Sea, 1974 (SOLAS 74), which carries more than 12 passengers;
- (33) ‘light-duty vehicle’ means a Category M₁ motor vehicle as described in Article 4(1), point (a)(i) or a Category N₁ motor vehicle as described in Article 4(1), point (b)(i), of Regulation (EU) 2018/858;
- (34) ‘liquefied methane’ means LNG, liquefied biogas or synthetic liquefied methane, including blends of those fuels;

⁽³⁾ Regulation (EU) 2015/757 of the European Parliament and of the Council of 29 April 2015 on the monitoring, reporting and verification of greenhouse gas emissions from maritime transport, and amending Directive 2009/16/EC (OJ L 123, 19.5.2015, p. 55).

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- (35) ‘manufacturer’ means a ‘manufacturer’ as defined in Article 3, point (40), of Regulation (EU) 2018/858;
- (36) ‘mobility service provider’ means a legal person that provides services in return for remuneration to an end user, including the selling of recharging or refuelling services;
- (37) ‘normal power recharging point’ means a recharging point with a power output less than or equal to 22 kW for the transfer of electricity to an electric vehicle;
- (38) ‘National Access Point’ means a digital interface set up by a Member State that constitutes a single point of access to data;
- (39) ‘operator of a recharging point’ means the entity that is responsible for the management and operation of a recharging point and that provides a recharging service to end users, including in the name and on behalf of a mobility service provider;
- (40) ‘operator of a refuelling point’ means the entity that is responsible for the management and operation of a refuelling point and that provides a refuelling service to end users, including in the name and on behalf of a mobility service provider;
- (41) ‘passenger ship’ means a ship which carries more than 12 passengers, including cruise ships, high-speed passenger crafts and ro-ro passenger ships;
- (42) ‘payment service’ means a ‘payment service’ as defined in Article 4, point (3), of Directive (EU) 2015/2366 of the European Parliament and of the Council ⁽⁴⁾;
- (43) ‘plug-in hybrid vehicle’ means an electric vehicle with a conventional combustion engine combined with an electric propulsion system which can be recharged from an external electric power source;
- (44) ‘power output’ means the theoretical maximum power, expressed in kW, that a recharging point, station or pool, or a shore-side electricity supply installation can provide to vehicles or vessels connected to that recharging point, station, pool or installation;
- (45) ‘publicly accessible alternative fuels infrastructure’ means an alternative fuels infrastructure which is located at a site or premises that are open to the general public, irrespective of whether the alternative fuels infrastructure is located on public or private property, whether limitations or conditions apply in terms of access to the site or premise and irrespective of the applicable use conditions of the alternative fuels infrastructure;
- (46) ‘Quick Response code’ (QR code) means an ISO/IEC 18004:2015-compliant encoding and visualisation of data;

⁽⁴⁾ Directive (EU) 2015/2366 of the European Parliament and of the Council of 25 November 2015 on payment services in the internal market, amending Directives 2002/65/EC, 2009/110/EC and 2013/36/EU and Regulation (EU) No 1093/2010, and repealing Directive 2007/64/EC (OJ L 337, 23.12.2015, p. 35).

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- (47) ‘recharge on an ad hoc basis’ means a recharging service purchased by an end user without the need for that end user to register, conclude a written agreement or enter into a commercial relationship with the operator of that recharging point that goes beyond the mere purchase of the recharging service;
- (48) ‘recharging point’ means a fixed or mobile, on-grid or off-grid interface for the transfer of electricity to an electric vehicle which, although it may have one or more connectors to accommodate different connector types, is capable of recharging only one electric vehicle at a time, and which excludes devices with a power output less than or equal to 3,7 kW the primary purpose of which is not the recharging of electric vehicles;
- (49) ‘recharging point, station or pool dedicated to light-duty vehicles’ means a recharging point, station or pool intended for the recharging of light-duty vehicles, due to the specific design of the connectors/plugs or the design of the parking space adjacent to the recharging point, station or pool, or both;
- (50) ‘recharging point, station or pool dedicated to heavy-duty vehicles’ means a recharging point, station or pool intended for the recharging of heavy-duty vehicles, either due to the specific design of the connectors/plugs or to the design of the parking space adjacent to the recharging point, station or pool, or both;
- (51) ‘recharging pool’ means one or more recharging stations at a specific location;
- (52) ‘recharging station’ means a physical installation at a specific location, consisting of one or more recharging points;
- (53) ‘recharging service’ means the sale or provision of electricity, including related services, through a publicly accessible recharging point;
- (54) ‘recharging session’ means the full process of recharging a vehicle at a publicly accessible recharging point from the moment the vehicle is connected to the moment the vehicle is disconnected;
- (55) ‘refuel on an ad hoc basis’ means a refuelling service purchased by an end user without the need for that end user to register, conclude a written agreement, or enter into a commercial relationship with the operator of that refuelling point that goes beyond the mere purchase of the refuelling service;
- (56) ‘refuelling point’ means a refuelling facility for the provision of any liquid or gaseous fuel, through a fixed or a mobile installation, which is capable of refuelling only one vehicle, one train, one vessel or one aircraft at a time;
- (57) ‘refuelling service’ means the sale or provision of any liquid or gaseous fuel through a publicly accessible refuelling point;

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- (58) ‘refuelling session’ means the full process of refuelling a vehicle at a publicly accessible refuelling point from the moment the vehicle is connected to the moment the vehicle is disconnected;
- (59) ‘refuelling station’ means a single physical installation at a specific location, consisting of one or more refuelling points;
- (60) ‘regulatory authority’ means a regulatory authority designated by each Member State pursuant to Article 57(1) of Directive (EU) 2019/944;
- (61) ‘renewable energy’ means ‘energy from renewable sources’ as defined in Article 2, second paragraph, point (1), of Directive (EU) 2018/2001;
- (62) ‘ro-ro passenger ship’ means a ship with facilities to enable road or rail vehicles to roll on and roll off the vessel which carries more than 12 passengers;
- (63) ‘safe and secure parking area’ means a parking area accessible to drivers engaged in the carriage of goods or passengers which has been certified in accordance with Commission Delegated Regulation (EU) 2022/1012 ⁽⁵⁾;
- (64) ‘shore-side electricity supply’ means the provision of shore-side electrical power through a standardised fixed or mobile interface to seagoing ships or inland waterway vessels, moored at the quayside;
- (65) ‘smart recharging’ means a recharging operation in which the intensity of electricity delivered to the battery is adjusted in real-time, based on information received through electronic communication;
- (66) ‘static data’ means data that do not change often or on a regular basis;
- (67) ‘TEN-T comprehensive network’ means a comprehensive network within the meaning of Article 9 of Regulation (EU) No 1315/2013;
- (68) ‘TEN-T core network’ means a core network within the meaning of Article 38 of Regulation (EU) No 1315/2013;
- (69) ‘TEN-T core inland waterway port or TEN-T comprehensive inland waterway port’ means an inland waterway port of the TEN-T core network or TEN-T comprehensive network, as listed and categorised in Annex II to Regulation (EU) No 1315/2013;

⁽⁵⁾ Commission Delegated Regulation (EU) 2022/1012 of 7 April 2022 supplementing Regulation (EC) No 561/2006 of the European Parliament and of the Council with regard to the establishment of standards detailing the level of service and security of safe and secure parking areas and to the procedures for their certification (OJ L 170, 28.6.2022, p. 27).

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- (70) ‘TEN-T core maritime port or TEN-T comprehensive maritime port’ means a maritime port of the TEN-T core network or TEN-T comprehensive network, as listed and categorised in Annex II to Regulation (EU) No 1315/2013;
- (71) ‘transmission system operator’ means a ‘transmission system operator’ as defined in Article 2, point (35), of Directive (EU) 2019/944;
- (72) ‘urban node’ means an ‘urban node’ as defined in Article 3, point (p), of Regulation (EU) No 1315/2013.

*Article 3***Targets for recharging infrastructure dedicated to light-duty electric vehicles**

1. Member States shall ensure that, in their territory, publicly accessible recharging stations dedicated to light-duty electric vehicles are deployed in a way that is commensurate with the uptake of light-duty electric vehicles and that they provide sufficient power output for those vehicles.

To that end, Member States shall ensure that, at the end of each year, starting from 2024, the following power output targets are met cumulatively:

- (a) for each light-duty battery electric vehicle registered in their territory, a total power output of at least 1,3 kW is provided through publicly accessible recharging stations; and
- (b) for each light-duty plug-in hybrid vehicle registered in their territory, a total power output of at least 0,80 kW is provided through publicly accessible recharging stations.

2. When the share of light-duty battery electric vehicles compared to the total fleet of light-duty vehicles registered in the territory of a Member State reaches at least 15 % and the Member State demonstrates that the implementation of the requirements set out in paragraph 1, second subparagraph, has adverse effects in that Member State, in that it is discouraging private investments, and is no longer justified, that Member State may submit to the Commission a reasoned request for authorisation to apply lower requirements in terms of level of total power output or to cease to apply such requirements.

3. The Commission shall, within six months of receipt of a reasoned request submitted pursuant to paragraph 2, adopt a decision as justified in each case.

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4. Member States shall ensure a minimum coverage of publicly accessible recharging points dedicated to light-duty electric vehicles on the road network in their territory.

To that end, Member States shall ensure that:

(a) along the TEN-T core road network, publicly accessible recharging pools dedicated to light-duty electric vehicles and meeting the following requirements are deployed in each direction of travel with a maximum distance of 60 km between them:

(i) by 31 December 2025, each recharging pool offers a power output of at least 400 kW and includes at least one recharging point with an individual power output of at least 150 kW;

(ii) by 31 December 2027, each recharging pool offers a power output of at least 600 kW and includes at least two recharging points with an individual power output of at least 150 kW;

(b) along the TEN-T comprehensive road network, publicly accessible recharging pools dedicated to light-duty electric vehicles and meeting the following requirements are deployed in each direction of travel with a maximum distance of 60 km between them:

(i) by 31 December 2027, along at least 50 % of the length of the TEN-T comprehensive road network, each recharging pool offers a power output of at least 300 kW and includes at least one recharging point with an individual power output of at least 150 kW;

(ii) by 31 December 2030, each recharging pool offers a power output of at least 300 kW and includes at least one recharging point with an individual power output of at least 150 kW;

(iii) by 31 December 2035, each recharging pool offers a power output of at least 600 kW and includes at least two recharging points with an individual power output of at least 150 kW.

5. The calculation of the percentage of the length of TEN-T comprehensive road network referred to in paragraph 4, point (b)(i), shall be based on the following elements:

(a) for the calculation of the denominator: the total length of the TEN-T comprehensive road network within the territory of the Member State;

(b) for the calculation of the numerator: the cumulated length of the sections of the TEN-T comprehensive road network between two publicly accessible recharging pools dedicated to light-duty electric vehicles meeting the requirements set out in paragraph 4, point (b)(i), excluding any sections of the TEN-T comprehensive road network between two of those recharging pools that are more than 60 km apart.

6. A single publicly accessible recharging pool dedicated to light-duty electric vehicles may be deployed along the TEN-T road network for both directions of travel provided that:

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- (a) that recharging pool is easily accessible from both directions of travel;
- (b) that recharging pool is adequately signposted; and
- (c) the requirements set out in paragraph 4 in terms of the maximum distance between recharging pools, total power output of the recharging pool, number of recharging points and power output of single recharging points applicable for a single direction of travel are complied with for both directions of travel.

7. By way of derogation from paragraph 4 of this Article, along roads of the TEN-T network with total annual average daily traffic of fewer than 8 500 light-duty vehicles and where the deployment of infrastructure cannot be justified in socio-economic cost-benefit terms, Member States may provide that a publicly accessible recharging pool dedicated to light-duty electric vehicles serves both directions of travel provided that the requirements set out in paragraph 4 of this Article, in terms of the maximum distance between recharging pools, the total power output of the recharging pool, the number of recharging points and the power output of single recharging points applicable for a single direction of travel, are complied with, and that the recharging pool is easily accessible from both directions of travel and adequately signposted. Member States shall notify the Commission of any cases in which they have made use of the derogation referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

8. By way of derogation from paragraph 4 of this Article, along roads of the TEN-T network with a total annual average daily traffic of fewer than 8 500 light-duty vehicles and where the deployment of infrastructure cannot be justified in socio-economic cost-benefit terms, Member States may reduce, by up to 50 %, the total power output of a publicly accessible recharging pool dedicated to light-duty vehicles required pursuant to paragraph 4 of this Article, provided that that recharging pool serves only one direction of travel and that the other requirements set out in paragraph 4 of this Article, in terms of the maximum distance between recharging pools, the number of recharging points and the power output of single recharging points, are complied with. Member States shall notify the Commission of any cases in which they have made use of the derogation referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

9. By way of derogation from the requirement relating to the maximum distance of 60 km between the publicly accessible recharging pools dedicated to light-duty vehicles set out in paragraph 4, points (a) and (b), of this Article, Member States may allow a longer distance, of up to 100 km, for such recharging pools along roads of the TEN-T network with a total annual average daily traffic of fewer than 3 000 light-duty vehicles, provided that the distance between recharging pools is adequately signposted. Member States shall notify the Commission of any cases in which they have made use of the derogation referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

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10. Where a Member State has notified the Commission of a case in which it has made use of a derogation referred to in paragraph 7, the requirements set out in paragraph 4, points (a) and (b), in terms of the maximum distance between recharging pools shall be deemed to have been complied with.

11. Neighbouring Member States shall ensure that the maximum distances referred to in paragraph 4, points (a) and (b), are not exceeded for cross-border sections of the TEN-T core road network and TEN-T comprehensive road network.

*Article 4***Targets for recharging infrastructure dedicated to heavy-duty electric vehicles**

1. Member States shall ensure a minimum coverage of publicly accessible recharging points dedicated to heavy-duty electric vehicles in their territory.

To that end, Member States shall ensure that:

- (a) by 31 December 2025, along at least 15 % of the length of the TEN-T road network, publicly accessible recharging pools dedicated to heavy-duty electric vehicles are deployed in each direction of travel and that each recharging pool offers a power output of at least 1 400 kW and includes at least one recharging point with an individual power output of at least 350 kW;
- (b) by 31 December 2027, along at least 50 % of the length of the TEN-T road network, publicly accessible recharging pools dedicated to heavy-duty electric vehicles are deployed in each direction of travel and that each recharging pool:
 - (i) along the TEN-T core road network, offers a power output of at least 2 800 kW and includes at least two recharging points with an individual power output of at least 350 kW;
 - (ii) along the TEN-T comprehensive road network, offers a power output of at least 1 400 kW and includes at least one recharging point with an individual power output of at least 350 kW;
- (c) by 31 December 2030, along the TEN-T core road network, publicly accessible recharging pools dedicated to heavy-duty electric vehicles are deployed in each direction of travel with a maximum distance of 60 km between them and that each recharging pool offers a power output of at least 3 600 kW and includes at least two recharging points with an individual power output of at least 350 kW;
- (d) by 31 December 2030, along the TEN-T comprehensive road network, publicly accessible recharging pools dedicated to heavy-duty electric vehicles are deployed in each direction of travel with a maximum distance of 100 km between them and each recharging pool offers a power output of at least 1 500 kW and includes at least one recharging point with an individual power output of at least 350 kW;
- (e) by 31 December 2027, in each safe and secure parking area at least two publicly accessible recharging stations dedicated to heavy-duty electric vehicles with an individual power output of at least 100 kW are deployed;

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- (f) by 31 December 2030, in each safe and secure parking area at least four publicly accessible recharging stations dedicated to heavy-duty electric vehicles with an individual power output of at least 100 kW are deployed;
- (g) by 31 December 2025, in each urban node publicly accessible recharging points dedicated to heavy-duty electric vehicles with an aggregated power output of at least 900 kW are deployed, provided by recharging stations with an individual power output of at least 150 kW;
- (h) by 31 December 2030, in each urban node publicly accessible recharging points dedicated to heavy-duty electric vehicles with an aggregated power output of at least 1 800 kW are deployed, provided by recharging stations with an individual power output of at least 150 kW.

2. The calculation of the percentage of the length of TEN-T road network referred to in paragraph 1, points (a) and (b), shall be based on the following elements:

- (a) for the calculation of the denominator: the total length of the TEN-T road network within the territory of the Member State;
- (b) for the calculation of the numerator: the cumulated length of the sections of the TEN-T road network between two publicly accessible recharging pools dedicated to heavy-duty electric vehicles meeting the requirements set out in paragraph 1, points (a) or (b) respectively, excluding any sections of the TEN-T road network between two of those recharging pools that are more than 120 km apart.

3. A single publicly accessible recharging pool dedicated to heavy-duty electric vehicles may be deployed along the TEN-T road network for both directions of travel provided that:

- (a) that recharging pool is easily accessible from both directions of travel;
- (b) that recharging pool is adequately signposted; and
- (c) the requirements set out in paragraph 1 in terms of the maximum distance between recharging pools, total power output of the recharging pool, number of recharging points and power output of single points applicable for a single direction of travel are complied with for both directions of travel.

4. By way of derogation from paragraph 1 of this Article, along roads of the TEN-T network with a total annual average daily traffic of fewer than 2 000 heavy-duty vehicles and where the deployment of infrastructure cannot be justified in socio-economic cost-benefit terms, Member States may provide that a publicly accessible recharging pool dedicated to heavy-duty electric vehicles serves both directions of travel provided that the requirements set out in paragraph 1 of this Article, in terms of the maximum distance between recharging pools, the total power output of the recharging pool, the number of recharging points and the power output of single recharging points applicable for a single direction of travel, are complied with, and that the recharging pool is easily accessible from both directions of travel and adequately signposted. Member States shall notify the Commission of any cases in which they have made use of the derogation referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

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5. By way of derogation from paragraph 1 of this Article, along roads of the TEN-T network with a total annual average daily traffic of fewer than 2 000 heavy-duty vehicles and where the deployment of infrastructure cannot be justified in socio-economic cost-benefit terms, Member States may reduce by up to 50 % the total power output of a publicly accessible recharging pool dedicated to heavy-duty electric vehicles required pursuant to paragraph 1 of this Article, provided that that recharging pool serves only one direction of travel and that the other requirements set out in paragraph 1 of this Article, in terms of the maximum distance between recharging pools, the number of recharging points and the power output of single recharging points, are complied with. Member States shall notify the Commission of any cases in which they have made use of the derogation referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

6. By way of derogation from the requirement relating to the maximum distance of 60 km between the publicly accessible recharging pools dedicated to heavy-duty electric vehicles set out in paragraph 1, point (c), of this Article, Member States may allow a longer distance, of up to 100 km, for such recharging pools along roads of the TEN-T core network with a total annual average daily traffic of fewer than 800 heavy-duty vehicles, provided that the distance between recharging pools is adequately signposted. Member States shall notify the Commission of any cases in which they have made use of the derogations referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

7. Where a Member State has notified the Commission of a case in which it has made use of a derogation referred to in paragraph 6, the requirement set out in paragraph 1, point (c), in terms of the maximum distance between recharging pools, shall be deemed to have been complied with.

8. By way of derogation from the requirements set out in paragraph 1, points (a), (b), (c) and (d), relating to the total power output of publicly accessible recharging pools dedicated to heavy-duty electric vehicles and from the requirement set out in paragraph 1, point (c), relating to the maximum distance between those recharging pools, Cyprus may submit to the Commission a reasoned request for authorisation to apply lower requirements in terms of level of total power output of publicly accessible recharging pools dedicated to heavy-duty electric vehicles or to apply a longer maximum distance, of up to 100 km, between those recharging pools, or both, provided that such request, if authorised, would not impede the circulation of heavy-duty electric vehicles in that Member State.

The Commission shall, within six months of receipt of a reasoned request submitted pursuant to the first subparagraph, adopt a decision as justified in each case. Any authorisation granted to Cyprus pursuant to such decision shall be valid for a maximum of four years. Where Cyprus seeks to extend the validity of the authorisation, it may submit a further reasoned request to the Commission before the expiry of the authorisation.

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9. By 31 December 2030, neighbouring Member States shall ensure that the maximum distances between recharging pools referred to in paragraph 1, points (c) and (d), are not exceeded for cross-border sections of the TEN-T core road network and TEN-T comprehensive road network. Before that date, particular attention shall be paid to cross-border sections and neighbouring Member States shall make all possible efforts to respect those maximum distances as soon as they deploy the recharging infrastructure along the cross-border sections of the TEN-T road network.

*Article 5***Recharging infrastructure**

1. Operators of recharging points shall, at the publicly accessible recharging points operated by them, provide end users with the possibility to recharge their electric vehicle on an ad hoc basis.

At publicly accessible recharging points deployed from 13 April 2024, recharging on an ad hoc basis shall be possible using a payment instrument that is widely used in the Union. To that end, operators of recharging points shall accept electronic payments at those points through terminals and devices used for payment services, including at least one of the following:

- (a) payment card readers;
- (b) devices with a contactless functionality that is at least able to read payment cards;
- (c) for publicly accessible recharging points with a power output below 50 kW, devices using an internet connection and allowing for secure payment transactions such as those generating a specific Quick Response code.

From 1 January 2027, operators of recharging points shall ensure that all publicly accessible recharging points operated by them, with a power output equal to or more than 50 kW deployed along the TEN-T road network or deployed on a safe and secure parking area, including recharging points deployed before 13 April 2024, comply with the requirements set out in points (a) or (b).

A single payment terminal or device referred to in the second subparagraph may serve a number of publicly accessible recharging points within a recharging pool.

The requirements laid down in this paragraph shall not apply to publicly accessible recharging points that do not require payment for the recharging service.

2. Operators of recharging points shall ensure that when they offer automatic authentication at a publicly accessible recharging point operated by them, end users always have the right to not make use of the automatic authentication and instead either to recharge their vehicle on an ad hoc basis, as provided for in paragraph 1, or to use another contract-based recharging solution offered at that recharging point. Operators of recharging points shall clearly show that option to end users and offer it to them in a convenient manner at each publicly accessible recharging point operated by them, at which they make available automatic authentication.

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3. Prices charged by operators of publicly accessible recharging points shall be reasonable, easily and clearly comparable, transparent and non-discriminatory. Operators of publicly accessible recharging points shall not discriminate, through the prices charged, between end users and mobility service providers or between different mobility service providers. However, the level of prices may be differentiated, but only if the differentiation is proportionate and objectively justified.

4. At publicly accessible recharging points with a power output equal to or more than 50 kW, the ad hoc price charged by the operator shall be based on the price per kWh for the electricity delivered. In addition, the operators of those recharging points can charge an occupancy fee as a price per minute to discourage long occupancy of the recharging point.

Operators of publicly accessible recharging points with a power output equal to or more than 50 kW shall, at the recharging stations, show the ad hoc price per kWh and any possible occupancy fee expressed in price per minute so that that information is known to end users before they initiate a recharging session and price comparison is facilitated.

Operators of publicly accessible recharging points with a power output of less than 50 kW shall, at the recharging stations operated by them, make the information on the ad hoc price clearly and easily available, with all its price components, so that that information is known to end users before they initiate a recharging session and price comparison is facilitated. The applicable price components shall be presented in the following order:

- price per kWh;
- price per minute;
- price per session; and
- any other price component that applies.

The first and second subparagraphs shall apply to all recharging points deployed from 13 April 2024.

5. Prices charged by mobility service providers to end users shall be reasonable, transparent and non-discriminatory. Mobility service providers shall make available to end users, prior to the start of an intended recharging session, all price information specific to that recharging session, through freely available, widely supported electronic means, clearly distinguishing all price components, including applicable e-roaming costs and other fees or charges applied by the mobility service provider. The fees shall be reasonable, transparent and non-discriminatory. Mobility service providers shall not apply any extra charges for cross-border e-roaming.

6. Member States shall ensure that their authorities regularly monitor the recharging infrastructure market, and in particular, that they monitor the compliance of operators of recharging points and mobility service providers with paragraphs 3 and 5. Member States shall also seek to ensure that their authorities regularly monitor possibly unfair commercial practices affecting consumers.

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7. By 14 October 2024, operators of recharging points shall ensure that all publicly accessible recharging points operated by them are digitally-connected recharging points.

8. Operators of recharging points shall ensure that all publicly accessible recharging points operated by them and built after 13 April 2024 or renovated after 14 October 2024 are capable of smart recharging.

9. Member States shall take the necessary measures to ensure that within parking and rest areas along the TEN-T road network where alternative fuels infrastructure is deployed, the exact location of the alternative fuels infrastructure is adequately signposted.

10. By 14 April 2025, the operators of publicly accessible recharging points shall ensure that all direct current (DC) publicly accessible recharging points operated by them have a fixed recharging cable installed.

11. Where the operator of a recharging point is not the owner of that point, the owner shall make available to the operator, in accordance with the arrangements between them, a recharging point with the technical characteristics which enable the operator to comply with the obligations set out in paragraphs 2, 7, 8 and 10.

*Article 6***Targets for hydrogen refuelling infrastructure of road vehicles**

1. Member States shall ensure that, in their territory, a minimum number of publicly accessible hydrogen refuelling stations are deployed by 31 December 2030.

To that end, Member States shall ensure that by 31 December 2030 publicly accessible hydrogen refuelling stations designed for a minimum cumulative capacity of 1 tonne per day and equipped with at least a 700 bar dispenser are deployed with a maximum distance of 200 km between them along the TEN-T core network.

Member States shall ensure that, by 31 December 2030, at least one publicly accessible hydrogen refuelling station is deployed in each urban node. Member States shall ensure that an analysis is carried out to determine the best location for such refuelling stations and that the analysis examines in particular the deployment of such refuelling stations in multimodal hubs where other transport modes could also be supplied.

Member States shall set out in their national policy frameworks a clear linear trajectory towards meeting the 2030 targets, along with a clear indicative target for 2027 that delivers sufficient coverage of the TEN-T core network with a view to meeting developing market demands.

2. Neighbouring Member States shall ensure that the maximum distance referred to in paragraph 1, second subparagraph, is not exceeded for cross-border sections of the TEN-T core network.

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3. The operator of a publicly accessible refuelling station, or, where the operator is not the owner, the owner of that station in accordance with the arrangements between them, shall ensure that the station is designed to serve light-duty and heavy-duty vehicles.

4. By way of derogation from paragraph 1 of this Article, along roads of the TEN-T core network with a total annual average daily traffic of fewer than 2 000 heavy-duty vehicles and where the deployment of infrastructure cannot be justified in socio-economic cost-benefit terms, Member States may reduce by up to 50 % the capacity of a publicly accessible hydrogen refuelling station required pursuant to paragraph 1 of this Article, provided that the requirements set out in that paragraph in terms of the maximum distance between hydrogen refuelling stations and the dispenser pressure are complied with. Member States shall notify the Commission of any cases in which they have made use of the derogation referred to in this paragraph. Member States shall review those cases every two years as part of the national progress reporting referred to in Article 15.

5. By way of derogation from paragraph 1 of this Article, if the costs of the deployment of the infrastructure are disproportionate to the benefits, including the environmental benefits, Member States may decide not to apply paragraph 1 of this Article to:

- (a) outermost regions of the Union referred to in Article 349 TFEU; or
- (b) islands falling within the definition of small connected systems or small isolated systems according to Directive (EU) 2019/944.

In such cases, Member States shall justify their decisions to the Commission and shall make available all relevant information in their national policy frameworks.

Article 7

Hydrogen refuelling infrastructure

1. Operators of hydrogen refuelling points shall, at the publicly accessible refuelling points operated by them, provide end users with the possibility to refuel on an ad hoc basis.

Ad hoc refuelling shall be possible at all publicly accessible hydrogen refuelling points using a payment instrument that is widely used in the Union. To that end, operators of those points shall accept electronic payments through terminals and devices used for payment services, including at least one of the following:

- (a) payment card readers;
- (b) devices with a contactless functionality that is at least able to read payment cards.

For publicly accessible hydrogen refuelling points deployed after 13 April 2024, the requirements set out in this paragraph shall apply from their deployment. For publicly accessible refuelling points deployed before 13 April 2024, the requirements set out in this paragraph shall apply from 14 October 2024.

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Where the operator of the hydrogen refuelling point is not the owner of that point, the owner shall make available to the operator, in accordance with the arrangements between them, hydrogen refuelling points with the technical characteristics which enable the operator to comply with the obligations set out in this paragraph.

2. Prices charged by the operators of publicly accessible hydrogen refuelling points shall be reasonable, easily and clearly comparable, transparent and non-discriminatory. Operators of publicly accessible hydrogen refuelling points shall not discriminate, through the prices charged, between end users and mobility service providers or between different mobility service providers. However, the level of prices may be differentiated, but only if the differentiation is objectively justified.

3. Operators of hydrogen refuelling points shall clearly show information on the ad hoc price per kg at the publicly accessible hydrogen refuelling stations operated by them, so that that information is known to end users before they initiate a refuelling session and price comparison is facilitated.

4. Operators of publicly accessible hydrogen refuelling stations may provide hydrogen refuelling services to customers on a contractual basis, including in the name and on behalf of other mobility service providers. Mobility service providers shall charge prices to end users that are reasonable, transparent and non-discriminatory. Mobility service providers shall make available to end users all price information specific to their intended refuelling session prior to its start, through freely available, widely supported electronic means, clearly distinguishing all price components charged by the operator of the hydrogen refuelling point, including applicable e-roaming costs and other fees or charges applied by them.

*Article 8***Infrastructure for liquefied methane for road transport vehicles**

Until 31 December 2024, Member States shall ensure that an appropriate number of publicly accessible refuelling points for liquefied methane are deployed, at least along the TEN-T core network, in order to allow heavy-duty motor vehicles using liquefied methane to circulate throughout the Union, where there is demand, unless the costs of doing so are disproportionate to the benefits, including environmental benefits.

*Article 9***Targets for shore-side electricity supply in maritime ports**

1. Member States shall ensure that a minimum shore-side electricity supply for seagoing container ships and seagoing passenger ships is provided in TEN-T maritime ports.

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To that end, Member States shall take the necessary measures to ensure that by 31 December 2029:

- (a) TEN-T core maritime ports and TEN-T comprehensive maritime ports for which the annual number of port calls of ships that are moored at the quayside, averaged over the last three years, by seagoing container ships above 5 000 gross tonnes is above 100 are equipped to provide each year shore-side electricity supply for at least 90 % of the total number of port calls of seagoing container ships above 5 000 gross tonnes that are moored at the quayside at the maritime port concerned;
- (b) TEN-T core maritime ports and TEN-T comprehensive maritime ports for which the annual number of port calls of ships that are moored at the quayside, averaged over the last three years, by seagoing ro-ro passenger ships above 5 000 gross tonnes and seagoing high-speed passenger craft above 5 000 gross tonnes is above 40 are equipped to provide each year shore-side electricity supply for at least 90 % of the total number of port calls of seagoing ro-ro passenger ships above 5 000 gross tonnes and seagoing high-speed passenger craft above 5 000 gross tonnes that are moored at the quayside at the maritime port concerned;
- (c) TEN-T core maritime ports and TEN-T comprehensive maritime ports for which the annual number of port calls of ships that are moored at the quayside, averaged over the last three years, by seagoing passenger ships above 5 000 gross tonnes other than seagoing ro-ro passenger ships and seagoing high-speed passenger craft is above 25 are equipped to provide each year shore-side electricity supply for at least 90 % of the total number of port calls of seagoing passenger ships above 5 000 gross tonnes other than seagoing ro-ro passenger ships and seagoing high-speed passenger craft that are moored at the quayside at the maritime port concerned.

2. The port calls of ships referred to in Article 6(5), points (a), (b), (c), (e) and (g) of Regulation (EU) 2023/1805 shall not be taken into account for the purposes of determining the total number of port calls of ships that are moored at the quayside at the maritime port concerned under paragraph 1 of this Article.

3. Where the TEN-T core maritime port or TEN-T comprehensive maritime port is located on an island, or in an outermost region referred to in Article 349 TFEU or on the territory of Ceuta and Melilla, which is not connected directly to the electricity grid of the mainland, or in the case of an outermost region or of Ceuta and Melilla to the electricity grid of a neighbouring country, paragraph 1 of this Article shall not apply until such a connection has been completed or there is sufficient locally generated electricity capacity from non-fossil energy sources to cover the needs of the island, the outermost region or of Ceuta and Melilla, whichever is relevant.



Article 10

Targets for shore-side electricity supply in inland waterway ports

Member States shall ensure that:

- (a) at least one installation providing shore-side electricity supply to inland waterway vessels is deployed at all TEN-T core inland waterway ports by 31 December 2024;
- (b) at least one installation providing shore-side electricity supply to inland waterway vessels is deployed at all TEN-T comprehensive inland waterway ports by 31 December 2029.

Article 11

Targets for supply of liquefied methane in maritime ports

1. Member States shall ensure that an appropriate number of refuelling points for liquefied methane are deployed at TEN-T core maritime ports referred to in paragraph 2 to enable seagoing ships to circulate throughout the TEN-T core network by 31 December 2024. Member States shall cooperate with neighbouring Member States where that is necessary in order to ensure adequate coverage of the TEN-T core network.

2. Member States shall designate in their national policy frameworks TEN-T core maritime ports that provide access to the refuelling points for liquefied methane referred to in paragraph 1, taking into consideration port development, existing liquefied methane refuelling points and actual market demand, both in the short- and long-term, as well as other developments.

Article 12

Targets for supply of electricity to stationary aircraft

1. Member States shall ensure that, at all airports of the TEN-T core network and TEN-T comprehensive network, the provision of electricity supply to stationary aircraft is ensured as follows:

- (a) by 31 December 2024, at all aircraft contact stands used for commercial air transport operations to embark or disembark passengers or to load or unload goods;
- (b) by 31 December 2029, at all aircraft remote stands used for commercial air transport operations to embark or disembark passengers or to load or unload goods.

2. Member States may exempt airports of the TEN-T network with fewer than 10 000 commercial flight movements per year, averaged over the last three years, from the obligation to supply electricity to stationary aircraft at all aircraft remote stands.

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3. Paragraph 1 shall not apply to specially dedicated de-icing stands, stands inside designated military areas and stands specially dedicated to general aviation aircraft below 5,7 tonnes of maximum take-off weight.

4. As of 1 January 2030 at the latest, Member States shall take the necessary measures to ensure that the electricity supplied pursuant to paragraph 1 originates from the electricity grid or is generated on site without using fossil fuels.

*Article 13***Railway infrastructure**

As regards railway infrastructure that is not covered by Regulation (EU) No 1315/2013, Member States shall assess the development of alternative fuel technologies and propulsion systems for rail sections that cannot be fully electrified for technical or cost-efficiency reasons, such as hydrogen- or battery-powered trains, and, if relevant, any recharging and refuelling infrastructure needs.

*Article 14***National policy frameworks**

1. By 31 December 2024, each Member State shall prepare and transmit to the Commission a draft national policy framework for the development of the market as regards alternative fuels in the transport sector and the deployment of the relevant infrastructure.

2. The national policy framework shall contain at least the following elements:

- (a) an assessment of the current state and future development of the market as regards alternative fuels in the transport sector, and of the development of alternative fuels infrastructure, considering intermodal access of alternative fuels infrastructure and, where relevant, cross-border continuity and the development of alternative fuels infrastructure on islands and in the outermost regions;
- (b) national targets and objectives pursuant to Articles 3, 4, 6, 8, 9, 10, 11 and 12 for which mandatory national targets are set out in this Regulation;
- (c) policies and measures necessary to ensure that the mandatory targets and objectives referred to in point (b) are reached;
- (d) measures, planned or adopted, to promote the deployment of alternative fuels infrastructure for captive fleets, in particular for recharging stations and hydrogen refuelling stations for public transport services and recharging stations for car sharing;
- (e) measures, planned or adopted, to encourage and facilitate the deployment of recharging stations for light-duty and heavy-duty vehicles in private locations that are not accessible to the public;

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- (f) measures, planned or adopted, to promote alternative fuels infrastructure in urban nodes, in particular with respect to publicly accessible recharging points;
- (g) measures, planned or adopted, to promote a sufficient number of publicly accessible high-power recharging points;
- (h) measures, planned or adopted, necessary to ensure that the deployment and operation of recharging points, including the geographical distribution of bidirectional recharging points, contribute to the flexibility of the energy system and to the penetration of renewable electricity into the electric system;
- (i) measures to ensure that publicly accessible recharging and refuelling points for alternative fuels are accessible to older persons, persons with reduced mobility and persons with disabilities in accordance with the accessibility requirements of Directive (EU) 2019/882;
- (j) measures, planned or adopted, to remove possible obstacles with regard to planning, permitting, procuring and operating of alternative fuels infrastructure;
- (k) an overview of the state of play, perspectives and planned measures in respect of the deployment of alternative fuels infrastructure in maritime ports other than for liquefied methane and shore-side electricity supply for use by seagoing vessels, such as for hydrogen, ammonia, methanol and electricity;
- (l) an overview of the state of play, perspectives and planned measures in respect of deployment of alternative fuels infrastructure including targets, key milestones and financing needed, for hydrogen- or battery-powered trains on TEN-T rail sections that cannot be electrified;
- (m) an overview of the state of play, perspectives and planned measures in respect of deployment of alternative fuels infrastructure in airports other than for electricity supply to stationary aircraft, such as for electric recharging and hydrogen refuelling for aircrafts;
- (n) an overview of the state of play, perspectives, and planned measures in respect of deployment of alternative fuels infrastructure in inland navigation, such as for electricity and hydrogen.

3. The national policy framework may contain the following elements:

- (a) an overview of the state of play, perspectives and planned measures for the deployment of alternative fuels infrastructure in maritime ports, such as for electricity and hydrogen, for port services as defined in Regulation (EU) 2017/352 of the European Parliament and of the Council ⁽⁶⁾;

⁽⁶⁾ Regulation (EU) 2017/352 of the European Parliament and of the Council of 15 February 2017 establishing a framework for the provision of port services and common rules on the financial transparency of ports (OJ L 57, 3.3.2017, p. 1).

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- (b) national targets and measures to promote alternative fuels infrastructure along the road networks which are not included in the TEN-T core network or TEN-T comprehensive network, in particular with respect to publicly accessible recharging points;
 - (c) measures to guarantee accessibility of recharging and refuelling infrastructure in the entire territory of the Member State, paying particular attention to rural areas to ensure their accessibility and territorial cohesion;
 - (d) measures to ensure that the density of publicly accessible alternative fuels infrastructure available at national level takes into account population density;
 - (e) national targets and objectives for the deployment of alternative fuels infrastructure related to points (a), (b), (c) and (d) for which no mandatory targets are laid down in this Regulation.
4. Member States shall ensure that the national policy frameworks take into account the needs of the different transport modes existing on their territory.
5. Member States shall ensure that national policy frameworks take into account, as appropriate, the interests of regional and local authorities, in particular where recharging and refuelling infrastructure for public transport is concerned, as well as those of the stakeholders concerned.
6. Where necessary, Member States shall cooperate, by means of consultations or joint policy frameworks, to ensure that the measures required to achieve the objectives of this Regulation are coherent and coordinated. In particular, Member States shall cooperate on establishing strategies on the use of alternative fuels and on the deployment of corresponding infrastructure in waterborne transport. The Commission shall assist the Member States in the cooperation process.
7. Support measures for alternative fuels infrastructure shall comply with the relevant Union State aid rules.
8. Each Member State shall make its draft national policy framework publicly available and shall ensure that the public is given early and effective opportunities to participate in the preparation of the draft national policy framework.
9. The Commission shall assess the draft national policy frameworks and may issue recommendations to Member States. Those recommendations shall be issued no later than six months after the submission of the draft national policy frameworks as referred to in paragraph 1 of this Article. They may, in particular, address:
- (a) the level of ambition of targets and objectives with a view to complying with the obligations set out in Articles 3, 4, 6, 8, 9, 10, 11, 12 and 13;
 - (b) policies and measures relating to national targets and objectives.
10. Each Member State shall take due account of any recommendations from the Commission in its final national policy framework. If the Member State concerned does not address a recommendation or a substantial part thereof, that Member State shall provide a written explanation to the Commission.

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11. By 31 December 2025, each Member State shall draft its final national policy framework in an easily readable and understandable form and notify it to the Commission. Those final national policy frameworks shall be made publicly available by the Commission.

*Article 15***National reporting**

1. By 31 December 2027 and every two years thereafter, each Member State shall submit to the Commission a standalone national progress report on the implementation of its national policy framework. The report shall be drafted in an easily readable and understandable form and shall be made publicly available by the Commission.

2. The national progress report shall cover the information listed in Annex I and shall, where appropriate, include a relevant justification of the level of achievement of the national targets and objectives referred to in Article 14(2), as well as an indication of the measures to be taken to achieve those targets and objectives in the future.

3. By 30 June 2024 and every three years thereafter, Member States shall assess how the deployment and operation of recharging points could enable electric vehicles to further contribute to the flexibility of the energy system, including their participation in the balancing market, and to the further absorption of renewable electricity. That assessment shall take into account all types of recharging points, including those offering smart and bi-directional recharging, and all power outputs, whether public or private, and provide recommendations in terms of type of recharging point, supporting technology and geographical distribution in order to facilitate the ability of users to integrate their electric vehicles in the system. That assessment shall identify appropriate measures to be implemented in order to meet the requirements set out in this Regulation including those to ensure the consistency of infrastructure planning with the corresponding grid planning. That assessment shall take into account input from all stakeholders and shall be made publicly available. Each Member State may request its regulatory authority to carry out that assessment. On the basis of the results of the assessment, Member States shall, if necessary, take appropriate measures for the deployment of additional recharging points and include those measures in the national progress reports referred to in paragraph 1 of this Article. The assessment and measures shall be taken into account by the system operators in the network development plans referred to in Article 32(3) and Article 51 of Directive (EU) 2019/944.

4. On the basis of input from transmission system operators and distribution system operators, the regulatory authority of each Member State shall assess, by 30 June 2024 and every three years thereafter, the potential contribution of bidirectional recharging to reducing user and system costs and increasing the renewable electricity share in the electricity system. That assessment shall be made publicly available. On the basis of the results of the assessment, Member States shall, if necessary, take appropriate measures to adjust the availability and geographical distribution of bidirectional recharging points in private areas and include them in the national progress reports referred to in paragraph 1.

*Article 16***Content, structure and format of national policy frameworks and national progress reports**

By 14 October 2024, the Commission shall adopt guidance and templates concerning the content, structure and format of the national policy frameworks to be submitted by the Member States pursuant to Article 14 and the content of the national progress reports to be submitted by the Member States pursuant to Article 15(1). The Commission may adopt guidance and templates to facilitate the effective application across the Union of any other provisions of this Regulation.

*Article 17***Review of national policy frameworks and national progress reports**

1. By 31 December 2026, the Commission shall assess the national policy framework notified by Member States pursuant to Article 14(11) and shall submit to the European Parliament and to the Council a report on the assessment of those national policy frameworks and their coherence at Union level, including a first assessment of the expected level of achievement of the national targets and objectives referred to in Article 14(2).

2. The Commission shall assess the national progress reports submitted by Member States pursuant to Article 15(1) and shall, as appropriate, issue recommendations to Member States to ensure the achievement of the objectives and compliance with the obligations laid down in this Regulation.

3. The Member State concerned shall, within six months of receipt of the recommendations referred to in paragraph 2, notify the Commission of how it intends to implement the recommendations. If the Member State concerned decides not to implement the recommendations or a substantial part thereof, it shall provide the Commission with its reasons.

4. After the submission by the Member State of the notification or the reasoning referred to in paragraph 3, the Member State concerned shall set out in its next national progress report how it has implemented the recommendations.

5. The Commission shall submit to the European Parliament and to the Council a report on its assessment of the national progress reports one year after submission of those national progress reports by Member States pursuant to Article 15(1). That assessment shall contain an evaluation of:

- (a) the progress made by Member States with regard to the achievement of the targets and objectives referred to in Article 14(2), including the Member States' responses to the Commission recommendations pursuant to paragraph 2 of this Article;
- (b) the coherence of the development of alternative fuel infrastructure at Union level.

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6. On the basis of the final national policy frameworks referred to in Article 14(11), the national progress reports referred to in Article 15(1) and the reports referred to in Article 18(1), the Commission shall make publicly available and regularly update information on the national targets and the objectives submitted by each Member State regarding:

- (a) the number of publicly accessible recharging points and recharging stations, separately for recharging points dedicated to light-duty vehicles and recharging points and recharging stations dedicated to heavy-duty vehicles, and in accordance with the categorisation provided for in Annex III;
- (b) the number of publicly accessible hydrogen refuelling points;
- (c) the infrastructure for shore-side electricity supply in maritime ports and inland waterway ports of the TEN-T core network and TEN-T comprehensive network;
- (d) the infrastructure for electricity supply for stationary aircraft in airports of the TEN-T core network and TEN-T comprehensive network;
- (e) the number of refuelling points for liquefied methane in maritime ports and inland waterway ports of the TEN-T core network and TEN-T comprehensive network;
- (f) the number of publicly accessible refuelling points for liquefied methane for motor vehicles;
- (g) the number of publicly accessible refuelling points for CNG for motor vehicles;
- (h) recharging and refuelling points for other alternative fuels in maritime ports and inland waterway ports of the TEN-T core network and TEN-T comprehensive network;
- (i) recharging and refuelling points for other alternative fuels at airports of the TEN-T core network and TEN-T comprehensive network;
- (j) recharging and refuelling points for alternative fuels for rail transport.

Article 18

Progress tracking

1. By 31 March 2025 and by 31 March of every year thereafter, Member States shall report to the Commission the total aggregated recharging power output and the number of publicly accessible recharging points deployed and the number of battery electric vehicles and plug-in hybrid vehicles registered on their territory on 31 December of the previous year, in accordance with the requirements set out in Annex III.

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2. Without prejudice to the procedure laid down in Article 258 TFEU, where it is evident from the report referred to in paragraph 1 of this Article or from any information available to the Commission that a Member State is at risk of not achieving its national targets set out in Article 3(1) of this Regulation, the Commission may issue a finding to that effect and recommend the Member State concerned to take corrective measures to achieve the national targets. Within three months following the receipt of the Commission's findings, the Member State concerned shall notify to the Commission:

- (a) the corrective measures that it plans to implement in order to achieve the national targets set out in Article 3(1) of this Regulation, including any additional actions that the Member State intends to implement in order to achieve those targets; and
- (b) a clear timetable for actions that will enable the assessment of annual progress towards achievement of those targets.

Where the Commission finds that the corrective measures are satisfactory, the Member State concerned shall update its latest national progress report referred to in Article 15 with those corrective measures and submit it to the Commission.

The Commission shall make its recommendations and the corrective measures and additional actions by the Member State concerned publicly available.

Article 19

User information

1. Relevant, consistent and clear information shall be made available as regards motor vehicles placed on the market which are capable of being regularly recharged or refuelled.

That information shall be made available:

- (a) in motor vehicle manuals and on motor vehicles, by the manufacturers, when those vehicles are placed on the market;
- (b) at recharging and refuelling points, by recharging and refuelling point operators; and
- (c) in motor vehicle dealerships, by the distributors.

2. Whether vehicles and infrastructures or fuels and vehicles covered by paragraph 1 of this Article are compatible shall be determined in compliance with the technical specifications referred to in points 10.1 and 10.2 of Annex II.

Where such technical specifications refer to a graphical expression, including a colour coding scheme, the graphical expression shall be simple and easy to understand.

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That graphical expression shall be placed in a clearly visible manner:

- (a) by refuelling point operators, on corresponding pumps and their nozzles at all refuelling points operated by them, from the date on which fuels are placed on the market;
- (b) by manufacturer, in the immediate proximity of all fuel tanks' filling caps of motor vehicles recommended for and compatible with that fuel and in motor vehicle manuals, when such motor vehicles are placed on the market.

3. When fuel prices are shown at a refuelling station, Member States shall ensure that a comparison of the relevant unit prices is shown, where appropriate, and in particular for hydrogen, for information purposes following the common methodology for alternative fuels unit price comparison referred to in point 10.3 of Annex II.

4. In situations where European standards setting the technical specifications of a fuel do not include labelling requirements to indicate compliance with the standards concerned, where the labelling requirements do not refer to a graphical expression, including colour coding schemes, or where the labelling requirements are not suitable for achieving the objectives of this Regulation, the Commission may, for the purposes of the uniform implementation of paragraphs 1 and 2, mandate the European standardisation organisations to develop compatibility labelling specifications.

On the basis of the compatibility labelling specifications developed by the European standardisation organisations pursuant to the mandate referred to in the first subparagraph, the Commission shall adopt implementing acts determining the graphical expression, including a colour coding scheme, of compatibility for fuels introduced in the Union market which reach the level of 1 % of the total volume of sales, in the assessment of the Commission, in more than one Member State.

Those implementing acts shall be adopted in accordance with the examination procedure referred to in Article 23(2).

5. Where provisions on labelling of the respective European standards are updated or new European standards for alternative fuels are developed, the corresponding requirements on labelling shall apply to all recharging and refuelling points no later than 24 months after the corresponding implementing act is adopted and to all motor vehicles placed on the market from the date that the corresponding implementing act enters into force.

Article 20

Data provisions

1. Member States shall appoint an Identification Registration Organisation ('IDRO'). The IDRO shall issue and manage unique identification ('ID') codes to identify at least operators of recharging points and mobility service providers, by 14 April 2025.

▼ M2

2. By 14 April 2025, operators of publicly accessible recharging points and refuelling points for alternative fuels, or, in accordance with the arrangements between them, the owners of those points, shall ensure the availability of static data and dynamic data concerning alternative fuels infrastructure operated by them, or services inherently linked to such infrastructure that they provide or they outsource, at no cost. The following data types shall be made available:

(a) Static data for publicly accessible recharging and refuelling infrastructure for alternative fuels operated by them:

- (i) Legal name of the recharging or refuelling point operator or owner,
- (ii) Commercial name of the recharging or refuelling point operator or owner,
- (iii) Number of recharging or refuelling points,
- (iv) Service support,
- (v) Helpdesk telephone,
- (vi) Facilities offering associated services to the user,
- (vii) Global navigation satellite system (GNSS) geographic location information,
- (viii) Additional geographic location information,
- (ix) Country,
- (x) Region,
- (xi) City or town,
- (xii) Postal code,
- (xiii) Address name,
- (xiv) Opening time,
- (xv) Time zone,
- (xvi) Vehicle-type compatibility,
- (xvii) Vehicle specifications permitted,
- (xviii) Number of parking spaces,
- (xix) Number of parking spaces for people with disabilities,
- (xx) Payment device with bank card reader,
- (xxi) Payment device with a contactless functionality that is at least able to read payment cards,
- (xxii) Other ad hoc payment option,
- (xxiii) Additional information about payment providers accepted,
- (xxiv) Contract-based (subscription) payment option;

▼ M2

- (b) Further static data for publicly accessible recharging infrastructure operated by them:
 - (i) Recharging point ID code (connector),
 - (ii) Number of connectors,
 - (iii) Type of connector (plug),
 - (iv) Type of current,
 - (v) Recharging station maximum power,
 - (vi) Recharging point maximum power,
 - (vii) Mobility service providers offering contract-based recharging,
 - (viii) Plug-and-charge,
 - (ix) Smart recharging services,
 - (x) Electricity supplied is 100 % renewable;
- (c) Further static data for publicly accessible hydrogen refuelling infrastructure operated by them:
 - (i) Hydrogen state,
 - (ii) Hydrogen pressure,
 - (iii) Daily cumulative capacity,
 - (iv) Hydrogen supplied is 100 % renewable;
- (d) Further static data for publicly accessible refuelling infrastructure for liquified methane:
 - (i) Liquified methane supplied is 100 % renewable;
- (e) Further static data for publicly accessible refuelling infrastructure for alternative fuels operated by them:
 - (i) Type of connector (dispenser);
- (f) Dynamic data for publicly accessible recharging and refuelling infrastructure for alternative fuels operated by them:
 - (i) Operational status,
 - (ii) Availability,
 - (iii) Ad hoc price;
- (g) Further dynamic data for publicly accessible hydrogen refuelling infrastructure operated by them:
 - (i) Limited amount of hydrogen available

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3. Each operator of publicly accessible recharging and refuelling points for alternative fuels, or, in accordance with the arrangements between them, the owner of those points, shall set up an Application Programme Interface (API) that provides free and unrestricted access to the data referred to in paragraph 2, and shall submit information on that API to the national access points.

The API of each operator of recharging and refuelling points, or, in accordance with the arrangements between them, the API of the owner of those points, shall comply with common technical requirements established by the Commission in the delegated acts referred to in paragraph 6 to enable an automated and uniform data exchange between the operators of publicly accessible recharging and refuelling points and data users.

4. By 31 December 2024, Member States shall ensure that the data referred to in paragraph 2 of this Article are made accessible on an open and non-discriminatory basis to all data users through their national access points in accordance with the relevant provisions related to such data in Delegated Regulation (EU) 2022/670 and in compliance with the additional complementary specifications that may be adopted in accordance with paragraph 7 of this Article. Where Member States aggregate data under their national access points, they may provide those data to a common European access point by means of an API.

5. By 31 December 2026, the Commission shall establish a common European access point to function as a data gateway facilitating the access to the data referred to in paragraph 2 from the different national access points. The Commission shall ensure that the common European access point is easily accessible and can be used by all data users, for example through the creation of a dedicated web portal.

6. The Commission shall be empowered to adopt delegated acts in accordance with Article 22 to:

- (a) amend paragraph 2 of this Article to include additional data types concerning publicly accessible recharging points and refuelling points for alternative fuels or services inherently linked to such infrastructure that the operators of that infrastructure provide or outsource in view of technological developments or new services made available on the market; and
- (b) supplement this Regulation by laying down common technical requirements for a common application programme interface to enable an automated and uniform data exchange between the operators of publicly accessible recharging points and refuelling points for alternative fuels and data users.

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7. The Commission may adopt implementing acts laying down:

- (a) specifications that are complementary to those set out in Delegated Regulation (EU) 2022/670, related to the data format, frequency and quality in which the data referred to in paragraph 2 of this Article and in the delegated acts adopted on the basis of paragraph 6 of this Article shall be made available;
- (b) detailed procedures enabling the availability and accessibility of data required pursuant to this Article.

Those implementing acts shall be adopted in accordance with the examination procedure referred to in Article 23(2).

Those implementing acts shall be without prejudice to Directive 2010/40/EU of the European Parliament and of the Council ⁽⁷⁾ and the delegated and implementing acts adopted on the basis thereof.

8. The delegated and implementing acts referred to in paragraphs 6 and 7 shall provide for reasonable transitional periods before the provisions contained therein, or amendments thereof, become binding on the operators or owners of recharging points and refuelling points for alternative fuels.

Article 21

Common technical specifications

1. The technical specifications set out in Annex II shall apply.

2. In accordance with Article 10 of Regulation (EU) No 1025/2012, the Commission may request European standardisation organisations to draft European standards setting technical specifications for areas referred to in Annex II to this Regulation for which no common technical specifications have been adopted by the Commission.

3. The Commission shall adopt delegated acts in accordance with Article 22 to:

- (a) amend Annex II by introducing technical specifications for the areas listed in that Annex to enable full technical interoperability of the recharging and refuelling infrastructure in terms of physical connections, communication exchanges and access for persons with reduced mobility to those areas; and
- (b) without undue delay and at the latest 12 months after the adoption of the relevant standards, amend Annex II by updating the references to the standards referred to in the technical specifications set out in that Annex.

⁽⁷⁾ Directive 2010/40/EU of the European Parliament and of the Council of 7 July 2010 on the framework for the deployment of Intelligent Transport Systems in the field of road transport and for interfaces with other modes of transport (OJ L 207, 6.8.2010, p. 1).

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4. Where delegated acts referred to in paragraph 3 are to apply to existing infrastructures, those delegated acts shall be based on a cost-benefit analysis to be submitted to the European Parliament and the Council together with those delegated acts.

5. Where European standards setting the technical specifications of a fuel are developed after the adoption by the Commission of an implementing act referred to in Article 19(4), second subparagraph, and they include provisions requiring labelling to indicate compliance with the standards concerned and refer to a graphical expression, including colour coding schemes, the amendments of Annex II adopted by the delegated acts referred to in paragraph 3 of this Article shall include an indication of which of those standards or implementing acts are to apply and, where appropriate, repeal the relevant implementing acts.

6. The amendments of Annex II adopted by the delegated acts referred to in paragraph 3 shall include reasonable transitional periods for any technical specifications that those delegated acts introduce or amend, during which they shall not be binding in respect of the infrastructure concerned.

*Article 22***Exercise of the delegation**

1. The power to adopt delegated acts is conferred on the Commission subject to the conditions laid down in this Article.

2. The power to adopt delegated acts referred to in Articles 20 and 21 shall be conferred on the Commission for a period of five years from 13 April 2024. The Commission shall draw up a report in respect of the delegation of power not later than nine months before the end of the five-year period. The delegation of power shall be tacitly extended for periods of an identical duration, unless the European Parliament or the Council opposes such extension not later than three months before the end of each period.

3. The delegation of power referred in Articles 20 and 21 may be revoked at any time by the European Parliament or by the Council. A decision to revoke shall put an end to the delegation of the power specified in that decision. It shall take effect the day following the publication of the decision in the *Official Journal of the European Union* or at a later date specified therein. It shall not affect the validity of any delegated acts already in force.

4. Before adopting a delegated act, the Commission shall consult experts designated by each Member State in accordance with the principles laid down in the Interinstitutional Agreement of 13 April 2016 on Better Law-Making.

5. As soon as it adopts a delegated act, the Commission shall notify it simultaneously to the European Parliament and to the Council.

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6. A delegated act adopted pursuant to Articles 20 and 21 shall enter into force only if no objection has been expressed either by the European Parliament or the Council within a period of two months of notification of that act to the European Parliament and the Council or if, before the expiry of that period, the European Parliament and the Council have both informed the Commission that they will not object. That period shall be extended by three months at the initiative of the European Parliament or of the Council.

*Article 23***Committee procedure**

1. The Commission shall be assisted by a committee. That committee shall be a committee within the meaning of Regulation (EU) No 182/2011.

2. Where reference is made to this paragraph, Article 5 of Regulation (EU) No 182/2011 shall apply.

Where the committee delivers no opinion, the Commission shall not adopt the draft implementing act and of Article 5(4), third subparagraph, of Regulation (EU) No 182/2011 shall apply.

*Article 24***Reporting and review**

1. By 31 December 2024, the Commission shall submit to the European Parliament and to the Council a technology and market-readiness report dedicated to heavy-duty vehicles. That report shall take into account the initial indications of the preferences of the market. It shall also consider technological developments and the development of the technical specifications achieved by that date and developments expected in the short term, in particular regarding recharging and refuelling standards and technologies, such as high-power recharging standards and electric road systems, and the use of liquid hydrogen.

Regarding hydrogen refuelling stations, the Commission shall further assess the requirements referred to in Article 6 in light of the technological and market developments, the need to specify a higher capacity for those stations, the need to specify targets for liquid hydrogen refuelling infrastructure, as well as the date for the extension of the requirements for the deployment of hydrogen refuelling stations to the TEN-T comprehensive network.

2. By 31 December 2026 and every five years thereafter, the Commission shall review this Regulation.

In its review, the Commission shall assess, in particular, the following elements:

- (a) whether the traffic thresholds referred to in Article 3(6) and (7), in Article 4(4) and (5) and in Article 6(4) are still relevant in view of the expected increase of the share of hydrogen-powered vehicles or battery electric vehicles compared to the total fleet of vehicles circulating in the Union;

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- (b) whether the electronic means of payment referred to in Article 5(1) are still appropriate;
- (c) the functioning of the pricing mechanism for publicly accessible recharging stations and whether the pricing components laid down in Article 5(4), provide consumers with clear and sufficient information;
- (d) a possible decrease of the gross tonnage threshold, laid down in Article 9, as well as a possible extension of the scope of this Regulation to other ship types following relevant adjustments in other relevant Union legal acts;
- (e) the current state and future development of the market for hydrogen and electric propulsion aviation;
- (f) the effects of this Regulation as regards the potential and the magnitude of carbon leakage.

As part of this review, the Commission shall also assess the extent to which the implementation of this Regulation has met its objectives and the extent to which it has impacted the competitiveness of the relevant sectors covered by it. That review shall also cover the interaction of this Regulation with other relevant Union legal acts and shall identify any provisions that could be updated and simplified, as well as actions and measures that have been or could be taken to reduce the total cost pressure on relevant sectors. As part of the Commission's analysis of the efficiency of this Regulation, the review shall also include an assessment of the burden this Regulation imposes on businesses.

3. The Commission shall consider, if appropriate, whether to accompany that review with a proposal to amend this Regulation, in view of the outcome of the assessment referred to in paragraph 2.

*Article 25***Repeal**

1. Directive 2014/94/EU and Delegated Regulations (EU) 2019/1745 and (EU) 2021/1444 are repealed with effect from 13 April 2024.
2. References to Directive 2014/94/EU shall be construed as references to this Regulation and shall be read in accordance with the correlation table in Annex IV.

*Article 26***Entry into force and application**

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

It shall apply from 13 April 2024.

This Regulation shall be binding in its entirety and directly applicable in all Member States.



ANNEX I

Reporting

The national progress report referred to in Article 15(1) shall include at least the following elements:

1. target setting

(a) vehicle uptake projections for 31 December of the years 2025, 2030 and 2035 for:

- light-duty vehicles, separately for battery electric light-duty vehicles, plug-in hybrid light-duty vehicles and hydrogen-powered light-duty vehicles;
- heavy-duty vehicles, separately for battery electric heavy-duty vehicles and hydrogen-powered heavy-duty vehicles;

(b) targets for 31 December of the years 2025, 2027, 2030 and 2035 for:

- recharging infrastructure dedicated to light-duty electric vehicles: number of recharging stations and power output (classification of recharging stations in accordance with Annex III);
- development of recharging stations for light-duty electric vehicles not accessible to the public, if applicable;
- recharging infrastructure dedicated to heavy-duty electric vehicles: number of recharging stations and power output;
- development of recharging stations for heavy-duty electric vehicles not accessible to the public, if applicable;
- hydrogen refuelling stations: number of refuelling stations, capacity of the refuelling stations and connector provided;
- road refuelling stations for liquefied methane: number of refuelling stations and capacity of stations;
- refuelling points for liquefied methane in maritime ports of the TEN-T core network and TEN-T comprehensive network, including location (port) and capacity per port;
- shore-side electricity supply in maritime ports of the TEN-T core network and TEN-T comprehensive network, including exact location (port) and capacity of each installation within the port;
- shore-side electricity supply in inland waterway ports of the TEN-T core network and TEN-T comprehensive network including location (port) and capacity;
- electricity supply for stationary aircraft, number of installations per airport of the TEN-T core network or airport of the TEN-T comprehensive network;
- other national targets and objectives for which no Union-wide mandatory national targets exist, if applicable. For alternative fuels infrastructure in ports, airports and for rail the location and capacity/size of the installation has to be reported;

2. utilisation rates: for the categories under point 1(b), reporting the utilisation of that infrastructure;

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3. the level of achievement of the national targets reported for the deployment of alternative fuels in the different transport modes (road, rail, water and air):
 - level of achievement of the infrastructure deployment targets as referred to in point 1(b) for all applicable transport modes, in particular for recharging stations, electric road system (if applicable), hydrogen refuelling stations, shore-side electricity supply in maritime ports and inland waterway ports, liquefied methane bunkering in TEN-T core maritime ports, other alternative fuels infrastructure in ports, electricity supply to stationary aircrafts;
 - for recharging points, specifying the ratio of public to private infrastructure;
 - deployment of alternative fuels infrastructure within urban nodes;
4. the review of the cases in which the Member States have made use of the derogations laid down in Article 3, paragraphs (6), (7) and (8), Article 4, paragraphs (6), (7) and (8) and Article 6, paragraph (4);
5. legal measures: information on legal measures, which may consist of legislative, regulatory or administrative measures to support the build-up of alternative fuels infrastructure, such as building permits, parking place permits, certification of the environmental performance of businesses and recharging and refuelling stations concessions;
6. information on the policy measures supporting the implementation of the national policy framework, including:
 - direct incentives for the purchase of means of transport that use alternative fuels or for building the infrastructure;
 - availability of tax incentives to promote means of transport that use alternative fuels and the relevant infrastructure;
 - use of public procurement in support of alternative fuels, including joint procurement;
 - demand-side non-financial incentives, for example preferential access to restricted areas, parking policy and dedicated lanes;
7. public deployment and manufacturing support, including:
 - annual public budget allocated for the deployment of alternative fuels infrastructure, broken down by alternative fuel and by transport mode (road, rail, water and air);
 - annual public budget allocated to support manufacturing plants for alternative fuel technologies, broken down by alternative fuel;
 - consideration of any particular needs during the initial phase of the deployment of alternative fuels infrastructure;
8. research, technological development and demonstration (RTD&D): annual public budget allocated to support alternative fuels RTD&D.

▼B*ANNEX II***Technical specifications****▼M1**

0. Definitions:

For the purpose of this Annex, the following definitions apply:

- (a) ‘installed’ means the initial placement of all relevant recharging point equipment, including hardware, software and associated electrical infrastructure, such as electricity supply connections, transformers, and other electrical systems, to enable the recharging of electric vehicles;
- (b) ‘renovated’ means a major or complete replacement of relevant recharging point equipment.

1.1. Normal-power recharging points for light-duty electric vehicles:

- alternating current (AC) normal-power recharging points for light-duty electric vehicles installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with socket-outlets or vehicle connectors of Type 2 for Mode 3 recharging as described in standard EN IEC 62196-2:2022 or, if their power is less than or equal to 3,7 kW and their primary purpose is the recharging of electric vehicles in Mode 2, with socket-outlets compliant with standard IEC 60884-1:2022; alternating current (AC) normal-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-2:2017 until they are renovated;
- direct current (DC) normal-power recharging points for light-duty electric vehicles installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with vehicle connectors of the combined charging system ‘Combo 2’ for Mode 4 recharging as described in standard EN IEC 62196-3:2022; direct current (DC) normal-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-3:2014 until they are renovated.

1.2. High-power recharging points for light-duty electric vehicles:

- alternating current (AC) high-power recharging points for light-duty electric vehicles installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with vehicle connectors of Type 2 for Mode 3 recharging as described in standard EN IEC 62196-2:2022; alternating current (AC) high-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-2:2017 until they are renovated;
- direct current (DC) high-power recharging points for light-duty electric vehicles installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with vehicle connectors of the combined charging system ‘Combo 2’ for Mode 4 recharging as described in standard EN IEC 62196-3:2022; direct current (DC) high-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-3:2014 until they are renovated.

▼ M1

1.3. Recharging points for L-category electric vehicles:

1.3.1. The publicly accessible alternating current (AC) recharging points reserved for L-category electric vehicles with a power output less than or equal to 3,7 kW shall be equipped, for interoperability purposes, with at least one of the following:

(a) socket-outlets or vehicle connectors of Type 3A as described in standard EN IEC 62196-2:2022 (for Mode 3 recharging);

(b) socket-outlets compliant with standard IEC 60884-1:2022 (for Mode 1 or Mode 2 recharging).

1.3.2. The publicly accessible alternating current (AC) recharging points reserved for L-category electric vehicles above 3,7 kW installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with socket-outlets or vehicle connectors of Type 2 for Mode 3 recharging as described in standard EN IEC 62196-2:2022; publicly accessible alternating current (AC) recharging points reserved for L-category electric vehicles above 3,7 kW installed before that date shall continue to comply with standard EN IEC 62196-2:2017 until they are renovated.

1.3.3. The publicly accessible direct current (DC) normal-power recharging points and high-power recharging points reserved for L-category electric vehicles installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with vehicle connectors of the combined charging system 'Combo 2' for Mode 4 recharging as described in standard EN IEC 62196-3:2022; publicly accessible direct current (DC) normal-power recharging points and high-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-3:2014 until they are renovated.

1.4. Normal power recharging points and high-power recharging points for electric buses:

— alternating current (AC) normal-power recharging points and high-power recharging points for electric buses that are installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with connectors of Type 2 for Mode 3 recharging as described in standard EN IEC 62196-2:2022; alternating current (AC) normal-power recharging points and high-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-2:2017 until they are renovated;

— direct current (DC) normal-power recharging points and high-power recharging points for electric buses that are installed or renovated from 8 January 2026 shall be equipped, for interoperability purposes, at least with vehicle connectors of the combined charging system 'Combo 2' for Mode 4 recharging as described in standard EN IEC 62196-3:2022; direct current (DC) normal-power recharging points and high-power recharging points installed before that date shall continue to comply with standard EN IEC 62196-3:2014 until they are renovated.

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- 1.5. Contact interface automated device for electric buses on conductive recharging in mode 4, in accordance with standard EN 61851-23-1:2020, shall be equipped at least with mechanical and electrical interfaces, as defined in the standard EN 50696:2021, concerning:

- automated connection device (ACD) mounted on the infrastructure (pantograph);
- automated connection device (ACD) mounted on the roof of the vehicle;
- automated connection device (ACD) mounted underneath the vehicle;
- automated connection device (ACD) mounted on the infrastructure and connecting to the side or on the roof of the vehicle.

▼M1

- 1.6. High-power recharging points for heavy-duty electric vehicles:

- Direct current (DC) high-power recharging points for recharging infrastructure capable of supplying electricity to both light- and heavy-duty electric vehicles shall be equipped, for interoperability purposes, at least with vehicle connectors of the combined charging system ‘Combo 2’ for Mode 4 recharging as described in standard EN IEC 62196-3:2022.

- 1.7. Technical specifications for inductive static wireless recharging for light-duty electric vehicles:

Recharging points for light-duty electric vehicles dedicated to inductive static wireless recharging shall comply, for interoperability purposes, with:

- EN IEC 61980-1:2021 ‘Electric vehicle wireless power transfer (WPT) systems – Part 1: General requirements’;
- EN IEC 61980-2:2023 ‘Electric vehicle wireless power transfer (WPT) systems – Part 2: Specific requirements for magnetic field wireless power transfer (MF-WPT) system communication and activities’;
- EN IEC 61980-3:2022 ‘Electric vehicle wireless power transfer (WPT) systems – Part 3: Specific requirements for magnetic field wireless power transfer (MF-WPT) systems’.

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- 1.8. Technical specifications for inductive static wireless recharging for heavy-duty electric vehicles.

- 1.9. Technical specifications for inductive dynamic wireless recharging for passenger cars and light-duty electric vehicles.

- 1.10. Technical specifications for inductive dynamic wireless recharging for heavy-duty-electric vehicles.

▼ B

- 1.11. Technical specifications for inductive static wireless recharging for electric buses.
- 1.12. Technical specifications for inductive dynamic wireless recharging for electric buses.
- 1.13. Technical specifications for electric road system for dynamic overhead power supply via a pantograph for heavy-duty electric vehicles.

▼ M1

- 1.14. Technical specifications for electric road system (ERS) for dynamic ground-level power supply through conductive rails for light- and heavy-duty electric vehicles:

Recharging infrastructure for alternating current (AC) and direct current (DC) dedicated to electric road system (ERS) for dynamic ground-level power supply through conductive rails for light- and heavy-duty electric vehicles equipped with ground level current collector devices, to enable conductive current collection by road vehicles from a feeding track integrated in the roadway shall comply, for interoperability purposes, with:

— CLC/TS 50717:2022 ‘Technical requirements for current collectors for ground-level feeding system on road vehicles in operation’.

▼ B

- 1.15. Technical specifications for battery swapping for L-category electric vehicles.
- 1.16. If technically feasible, technical specifications for battery swapping for passenger electric cars and light-duty electric vehicles.
- 1.17. If technically feasible, technical specifications for battery swapping for heavy-duty electric vehicles.
- 1.18. Technical specifications for recharging stations to ensure access to users with disabilities.
- 2. Technical specifications for communication exchange in the electric vehicle recharging sector

▼ M1

- 2.1. Technical specifications regarding communication between the electric vehicle and the recharging point (vehicle-to-grid communication):

- 2.1.1. The publicly accessible recharging points for alternating current (AC) and direct current (DC) for light- and heavy-duty electric vehicles installed or renovated from 8 January 2026 shall comply, for interoperability purposes, at least with the following standards:

— EN ISO 15118-1:2019 ‘Road vehicles – Vehicle to grid communication interface Part 1: General information and use-case definition’;

— EN ISO 15118-2:2016 ‘Road vehicles – Vehicle to grid communication Interface Part 2: Network and application protocol requirements’;

— EN ISO 15118-3:2016 ‘Road vehicles – Vehicle to grid communication interface Part 3: Physical and data link layer requirements’;

▼ M1

- EN ISO 15118-4:2019 ‘Road vehicles – Vehicle to grid communication interface Part 4: Network and application protocol conformance test’;
- EN ISO 15118-5:2019 ‘Road vehicles – Vehicle to grid communication interface – Part 5: Physical layer and data link layer conformance test’.

2.1.2. Publicly accessible recharging points for alternating current (AC) and direct current (DC) for light- and heavy-duty electric vehicles installed or renovated from 1 January 2027 shall comply, for interoperability purposes, at least with standard EN ISO 15118-20:2022 ‘Road vehicles – Vehicle-to-grid communication interface – Part 20: 2nd generation network layer and application layer requirements’. Where such recharging points offer automatic authentication and authorisation services, such as plug-and-charge, they shall comply, for interoperability and security purposes, with both standard EN ISO 15118-2:2016 and standard EN ISO 15118-20:2022.

2.1.3. Private recharging points for alternating current (AC) and direct current (DC) electric light- and heavy-duty electric vehicles installed or renovated from 1 January 2027 shall comply, for interoperability purposes, at least with the following standards:

- (a) EN IEC 61851-1:2019 ‘Electric vehicle conductive charging system – Part 1: General requirements’ (for Mode 2 recharging);
- (b) EN ISO 15118-20:2022 ‘Road vehicles – Vehicle to grid communication interface – Part 20: 2nd generation network layer and application layer requirements’ (for Mode 3 or Mode 4 recharging).

▼ B

- 2.2. Technical specifications regarding communication between the recharging point and the recharging point management system (back-end communication).
- 2.3. Technical specifications regarding communication between the recharging point operator, electromobility service providers and e-roaming platforms.
- 2.4. Technical specifications regarding communication between the recharging point operator and the distributed system operators.
- 3. Technical specifications for hydrogen supply for road transport vehicles

▼ M1

- 3.1. Technical specifications for connectors for refuelling points dispensing gaseous (compressed) hydrogen for light-duty vehicles shall comply, for interoperability purposes, at least with the interoperability requirements described in standard EN 17127:2024.

▼ B

- 3.2. The quality characteristics of hydrogen dispensed by hydrogen refuelling points for motor vehicles shall comply with the requirements described in standard EN 17124:2022. The methods to ensure that the hydrogen quality is met are also described in the standard.

▼ M1

- 3.3. The hydrogen refuelling algorithm shall comply with the requirements of standard EN 17127:2024.

▼B

- 3.4. Once the process of certification of standard EN ISO 17268:2020 is concluded, connectors for motor vehicles for the refuelling of gaseous hydrogen shall comply at least with that standard.

▼M1

- 3.5. Technical specifications for connectors for refuelling points dispensing gaseous (compressed) hydrogen for heavy-duty vehicles shall comply, for interoperability purposes, at least with the requirements described in standard EN 17127:2024.

▼B

- 3.6. Technical specifications for connectors for refuelling points dispensing liquefied hydrogen for heavy-duty vehicles.

- 4. Technical specifications for methane for road transport

- 4.1. Refuelling points for compressed natural gas (CNG) for motor vehicles shall comply with a fuelling pressure (service pressure) of 20,0 MPa gauge (200 bar) at 15 °C. A maximum fuelling pressure of 26,0 MPa with 'temperature compensation' is permitted in accordance with standard EN ISO 16923:2018.

- 4.2. The connector profile shall comply with Regulation No 110 of the Economic Commission for Europe of the United Nations referring to parts I and II in standard EN ISO 14469:2017.

- 4.3. Refuelling points for liquefied methane for motor vehicles shall comply with a fuelling pressure lower than the maximum allowable working pressure of the vehicle tank as addressed in standard EN ISO 16924:2018, 'Natural gas fuelling stations – LNG stations for fuelling vehicles'. In addition, the connector profile shall comply with standard EN ISO 12617:2017 'Road vehicles – Liquefied natural gas (LNG) refuelling connector – 3,1 MPa connector'.

- 5. Technical specifications for electricity supply for maritime transport and inland navigation

- 5.1. Shore-side electricity supply for seagoing ships, including the design, installation and testing of the systems, shall comply at least with the technical specifications of standard IEC/IEEE 80005-1:2019/AMD1:2022 for high-voltage shore connections.

- 5.2. Plugs, socket-outlets and ship couplers for high-voltage shore connection shall comply at least with the technical specification of standard IEC 62613-1:2019.

- 5.3. Shore-side electricity supply for inland waterway vessels shall comply at least with the standard EN 15869-2:2019 or standard EN 16840:2017 depending on energy requirements.

- 5.4. Technical specifications for shore-side battery recharging points for maritime vessels, featuring interconnectivity and system interoperability for maritime vessels.

- 5.5. Technical specifications for shore-side battery recharging points for inland navigation vessels, featuring interconnectivity and system interoperability for inland navigation vessels.

- 5.6. Technical specifications for vessel-to-port grid communication interfaces in automated on-shore power supply (OPS) and battery recharging systems for maritime vessels.

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- 5.7. Technical specifications for vessel-to-port grid communication interfaces in automated on-shore power supply (OPS) and battery recharging systems for inland navigation vessels.
- 5.8. If technically feasible, technical specifications for battery swapping and recharging at onshore stations for inland navigation vessels.
6. Technical specifications for hydrogen bunkering for maritime transport and inland navigation
- 6.1. Technical specifications for refuelling points and bunkering for gaseous (compressed) hydrogen for maritime hydrogen-powered vessels.
- 6.2. Technical specifications for refuelling points and bunkering for gaseous (compressed) hydrogen inland navigation hydrogen-powered vessels.
- 6.3. Technical specifications for refuelling points and bunkering for liquefied hydrogen for maritime hydrogen-powered vessels.
- 6.4. Technical specifications for refuelling points and bunkering for liquefied hydrogen inland navigation hydrogen-powered vessels.
7. Technical specifications for methanol bunkering for maritime transport and inland navigation
- 7.1. Technical specifications for refuelling points and bunkering for methanol for maritime methanol-fuelled vessels.
- 7.2. Technical specifications for refuelling points and bunkering for methanol for inland navigation methanol-fuelled vessels.
8. Technical specifications for ammonia bunkering for maritime transport and inland navigation
- 8.1. Technical specifications for refuelling points and bunkering for ammonia for maritime ammonia-fuelled vessels.
- 8.2. Technical specifications for refuelling points and bunkering for ammonia for inland navigation ammonia-fuelled vessels.
9. Technical specifications for liquefied methane refuelling points for maritime transport and inland navigation
- 9.1. Refuelling points for liquefied methane for seagoing ships which are not covered by the International Code of the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code) shall comply at least with standard EN ISO 20519:2017.
- 9.2. Refuelling points for liquefied methane for inland waterway vessels shall comply at least with standard EN ISO 20519:2017 (parts 5.3 to 5.7) for interoperability purposes only.

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10. Technical specifications related to fuel labelling
- 10.1. The ‘Fuels - Identification of vehicle compatibility - Graphical expression for consumer information’ label shall comply with standard EN 16942:2016+A1:2021.
- 10.2. The ‘Identification of vehicles and infrastructures compatibility - Graphical expression for consumer information on EV power supply’ shall comply at least with standard EN 17186:2019.
- 10.3. The common methodology for alternative fuels unit price comparison set out by Commission Implementing Regulation (EU) 2018/732 ⁽¹⁾.
- 10.4. Technical specifications for electric recharging stations and hydrogen refuelling facilities for rail transport.

⁽¹⁾ Commission Implementing Regulation (EU) 2018/732 of 17 May 2018 on a common methodology for alternative fuels unit price comparison in accordance with Directive 2014/94/EU of the European Parliament and of the Council (OJ L 123, 18.5.2018, p. 85).



ANNEX III

Reporting requirements on deployment of electric vehicles and publicly accessible recharging infrastructure

1. Member States must categorise their reporting on the deployment of electric vehicles as follows:
 - battery electric vehicles, separately for categories M₁, N₁, M_{2/3} and N_{2/3}
 - plug-in hybrid vehicles, separately for categories M₁, N₁, M_{2/3} and N_{2/3}
2. Member States must categorise their reporting on the deployment of publicly accessible recharging points as follows:

Category	Sub-category	Maximum power output	Definition pursuant to Article 2 of this Regulation
Category 1 (AC)	Slow AC recharging point, single-phase	$P < 7,4 \text{ kW}$	Normal-power recharging point
	Medium-speed AC recharging point, triple-phase	$7,4 \text{ kW} \leq P \leq 22 \text{ kW}$	
	Fast AC recharging point, triple-phase	$P > 22 \text{ kW}$	High-power recharging point
Category 2 (DC)	Slow DC recharging point	$P < 50 \text{ kW}$	
	Fast DC recharging point	$50 \text{ kW} \leq P < 150 \text{ kW}$	
	Level 1 - Ultra-fast DC recharging point	$150 \text{ kW} \leq P < 350 \text{ kW}$	
	Level 2 - Ultra-fast DC recharging point	$P \geq 350 \text{ kW}$	

3. The following data must be provided separately for publicly accessible recharging infrastructure dedicated to light-duty vehicles and heavy-duty vehicles:
 - number of recharging points, to be reported for each of the categories under point 2;
 - number of recharging stations, to be reported for each of the categories under point 2;
 - total aggregated power output of the recharging stations.



ANNEX IV

Correlation table

Directive 2014/94/EU	This Regulation
Article 1	Article 1
Article 2	Article 2
Article 3	Article 14
Article 4	Articles 3, 4, 5, 9 and 10
Article 5	Article 6
—	Article 7
Article 6	Articles 8 and 11
—	Article 12
—	Article 13
Article 7	Article 19
Article 8	Article 22
Article 9	Article 23
Article 10	Articles 15, 16 and 24
—	Article 17
—	Article 18
—	Article 20
—	Article 21
—	Article 25
Article 11	—
Article 12	Article 26
Article 13	—
Annex I	Annex I
Annex II	Annex II
—	Annex III